```
clc;
clear;
close all;

disp('⧫ Synthetic Genomic Dataset Generator');
```

```
⧫ Synthetic Genomic Dataset Generator
```

```
%% ========================
% PARAMETERS
%% ========================
numPatients = 50;     % number of patients
numGenes    = 10;     % number of genes

%% ========================
% Generate Patient IDs
%% ========================
PatientID = strings(numPatients,1);
for i = 1:numPatients
    PatientID(i) = "P" + sprintf('%03d', i);
end

%% ========================
% Generate Gene Names
%% ========================
GeneNames = strings(1,numGenes);
for j = 1:numGenes
    GeneNames(j) = "Gene" + j;
end

%% ========================
% Generate Gene Expression Data
%% ========================
% Simulated biological variation
GeneData = abs(randn(numPatients, numGenes) * 2 + 2);

%% ========================
% Generate Disease Labels
%% ========================
Disease = strings(numPatients,1);
for i = 1:numPatients
    if mean(GeneData(i,:)) > 2.2
        Disease(i) = "HighRisk";
    else
        Disease(i) = "LowRisk";
    end
end

%% ========================
```

```
% Create Table
%% ========================
T = table(PatientID);

for j = 1:numGenes
    T.(GeneNames(j)) = GeneData(:,j);
end

T.Disease = Disease;

%% ========================
% Save Dataset
%% ========================
filename = 'synthetic_genomic_data.csv';
writetable(T, filename);

disp('□ Synthetic Dataset Created Successfully!');
```

□ Synthetic Dataset Created Successfully!

```
disp(['□ File saved as: ', filename]);
```

□ File saved as: synthetic_genomic_data.csv

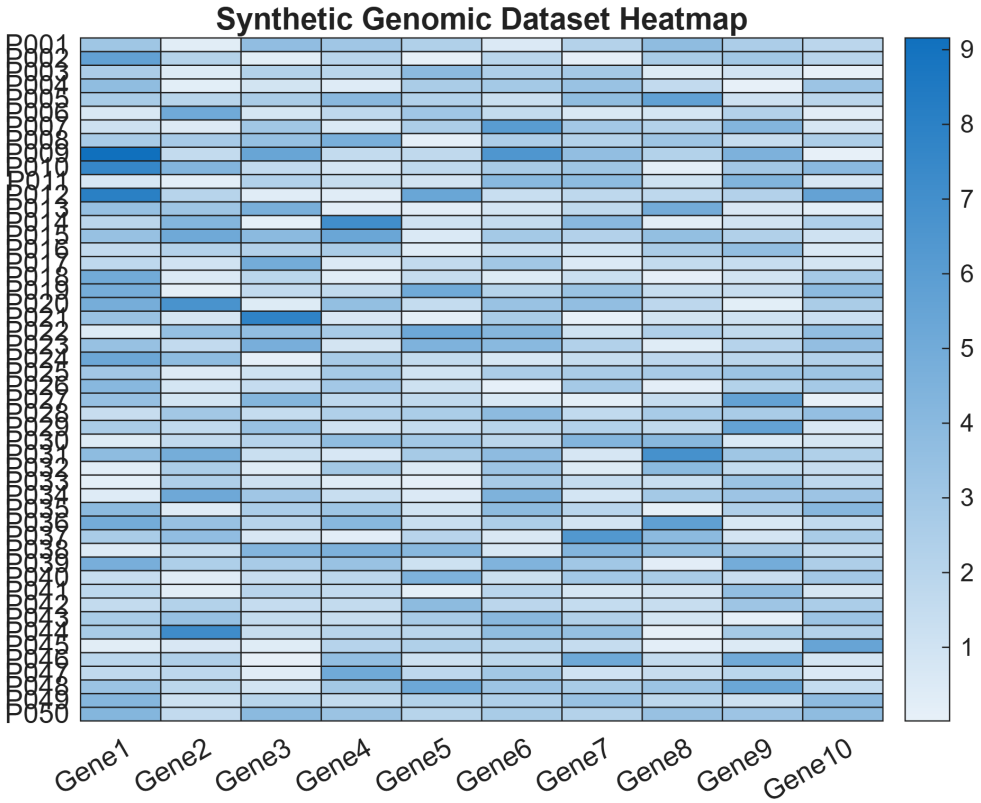
```
%% ========================
% Preview Data
%% ========================
disp('□ Dataset Preview:');
```

□ Dataset Preview:

```
disp(T(1:10,:));
```

PatientID	Gene1	Gene2	Gene3	Gene4	Gene5	Gene6	Gene7	Gene8
"P001"	3.0753	0.27269	3.6808	3.0401	2.3665	0.53166	2.1599	3.7619
"P002"	5.6678	2.1547	0.22394	1.9599	0.059535	1.9384	0.10304	2.6464
"P003"	2.5177	0.42823	2.2002	1.9305	3.8984	2.4647	2.823	0.43171
"P004"	3.7243	0.227	0.91094	0.40367	2.6141	2.8528	3.354	1.6107
"P005"	2.6375	1.9863	2.607	4.0374	2.2703	1.2544	3.7155	5.7172
"P006"	0.61538	5.0653	0.79935	1.7336	3.0305	1.5271	0.61768	0.79094
"P007"	1.1328	0.46067	2.9799	0.57094	2.5228	6.0474	2.8988	2.2067
"P008"	2.6852	2.7428	3.4787	4.7028	0.11703	2.5167	2.2013	3.1263
"P009"	9.1568	1.5488	5.4238	1.5505	1.6753	6.4589	3.6521	2.2272
"P010"	7.5389	4.2347	1.6118	0.82194	1.7079	2.6751	3.0723	0.19055

```
%% ========================
% Auto Plot Example
%% ========================
figure;
heatmap(GeneNames, PatientID, GeneData);
title('Synthetic Genomic Dataset Heatmap');
```


Synthetic Genomic Dataset Heatmap
Gene1
Gene2
Gene3
Gene4
Gene5
Gene6
Gene7
Gene8
Gene9
Gene10
9
8
7
6
5
4
3
2
1